

CSE 432 Robotics

Mobile Robots I

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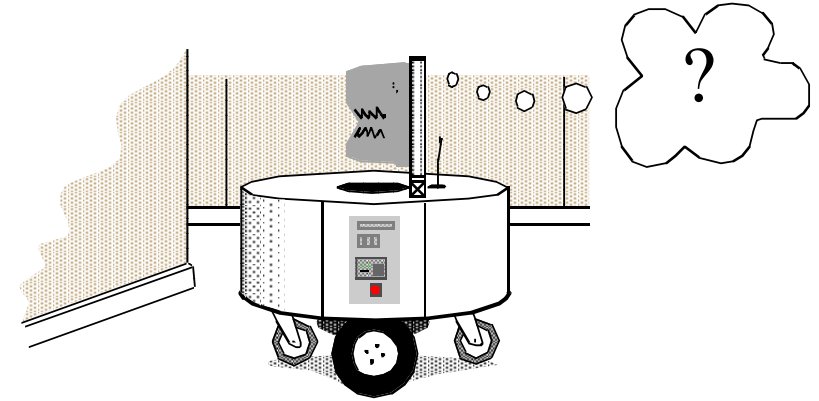
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Autonomous Mobile Robots

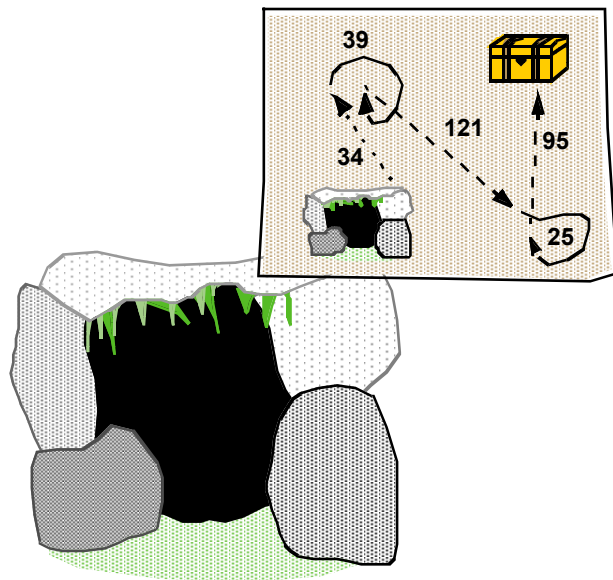
- The three key questions in Mobile Robotics
 - Where am I ?
 - Where am I going ?
 - How do I get there ?
- To answer these questions the robot has to
 - have a model of the environment (given or autonomously built)
 - perceive and analyze the environment
 - find its position within the environment
 - plan and execute the movement





Environment Representation and Modeling

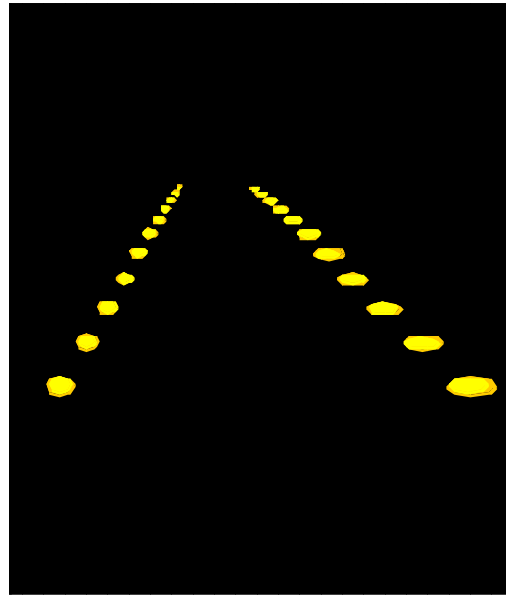
□ Odometry



How to find a treasure

➤ *not applicable*

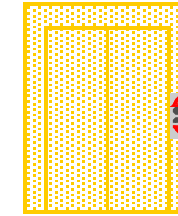
□ Modified Environments



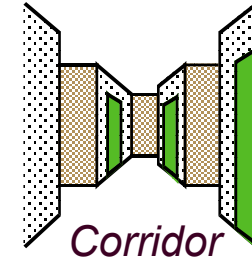
Landing at night

➤ *expensive, inflexible*

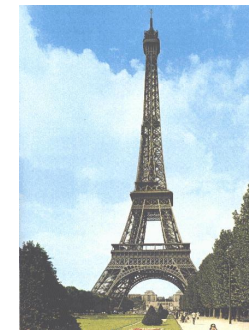
□ Feature-based Navigation



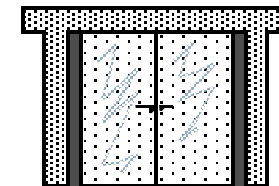
Elevator door



Corridor crossing



Eiffel Tower



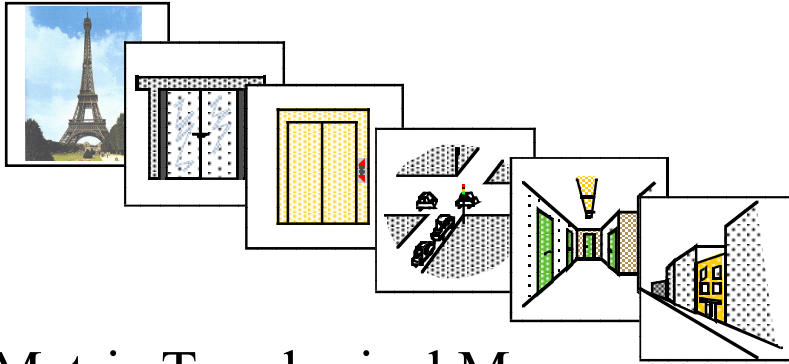
Entrance

➤ *still a challenge for artificial systems*

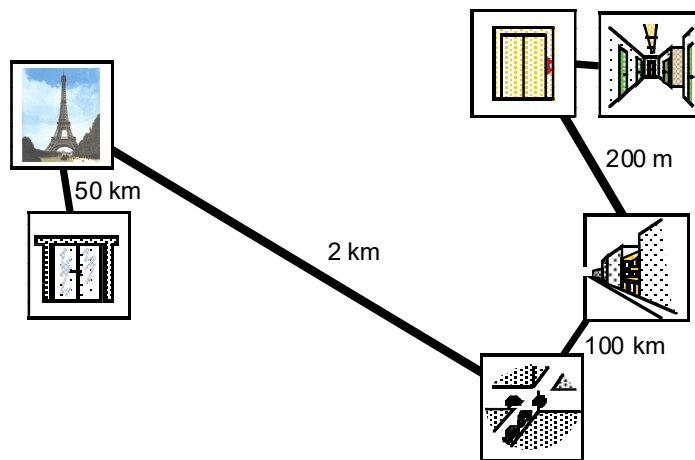


Environment Representation: The Map Categories

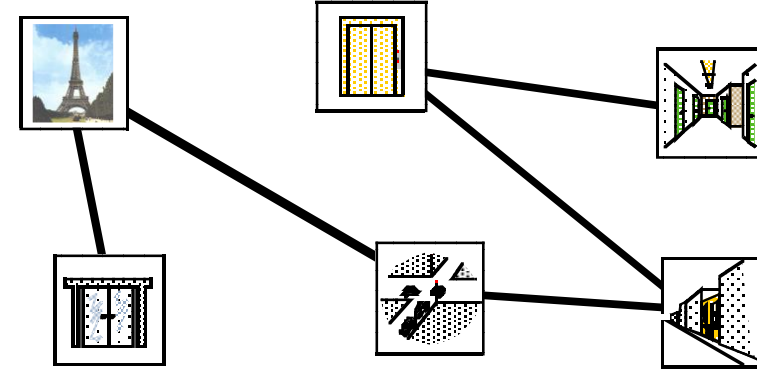
- Recognizable Locations



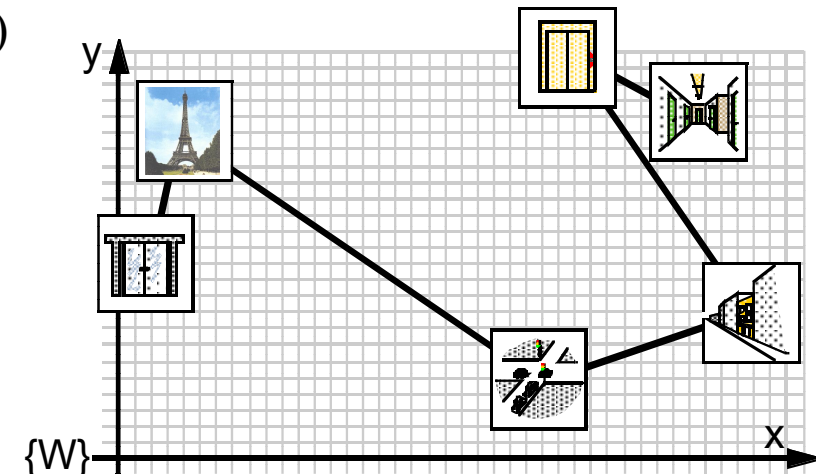
- ▢ Metric Topological Maps



- ▢ Topological Maps



- ▢ Fully Metric Maps (continuous or discrete)





Environment Models

- **Continuous**

- *position in x, y, θ*

- **Discrete**

- *metric grid*
- *topological grid*

- **Raw Data**

- *as perceived by sensor*

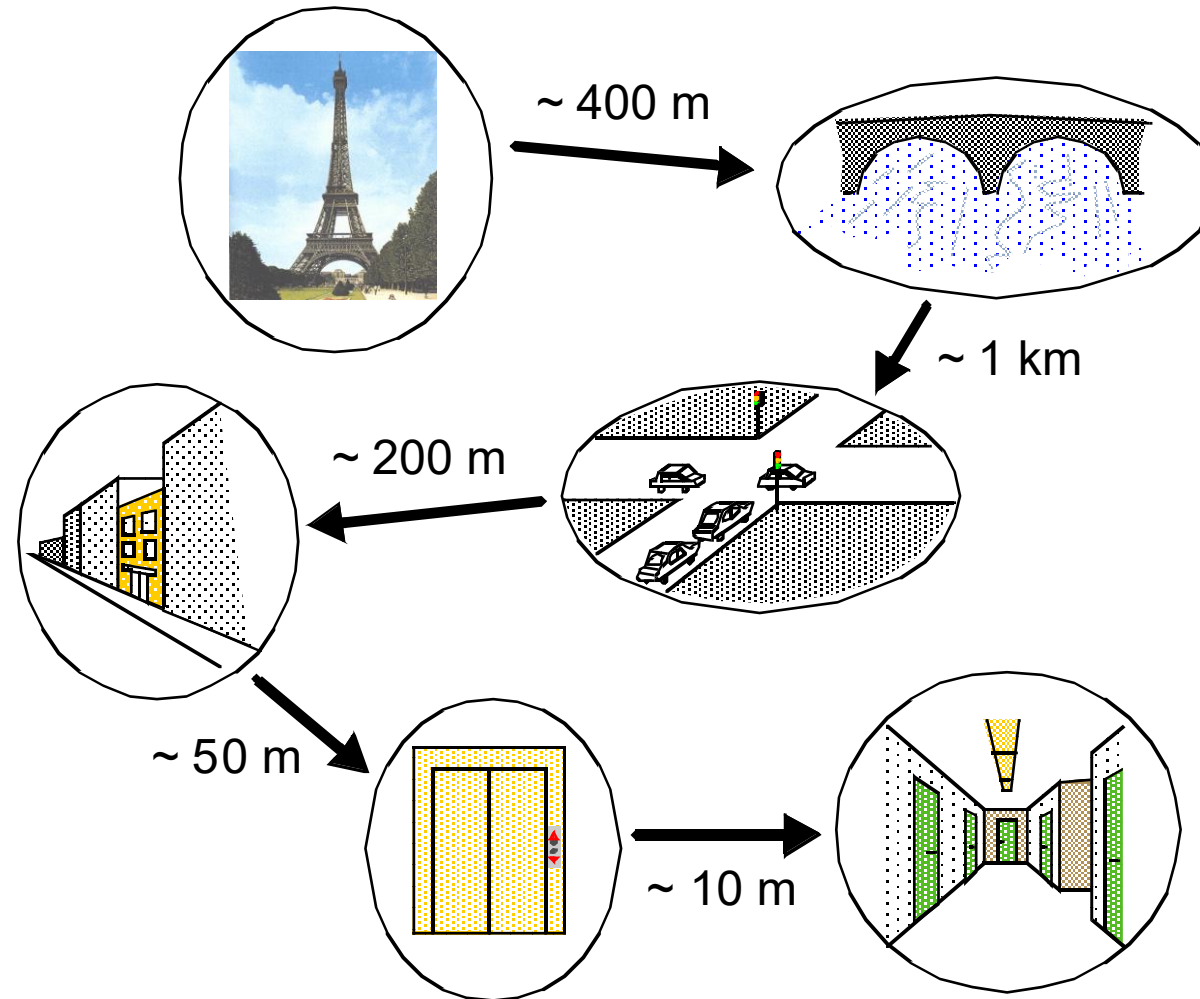
- **A feature** (or natural landmark) is an environmental structure which is static, always perceptible with the current sensor system and **locally unique**.

Examples

- *geometric elements (lines, walls, column ..)*
- a railway station
- a river
- the Eiffel Tower
- a human being
- fixed stars
- skyscraper



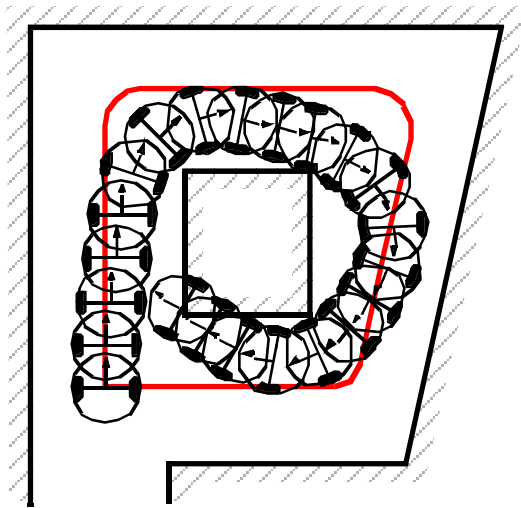
Human Navigation: Topological with imprecise metric information





Approaches with Limitations

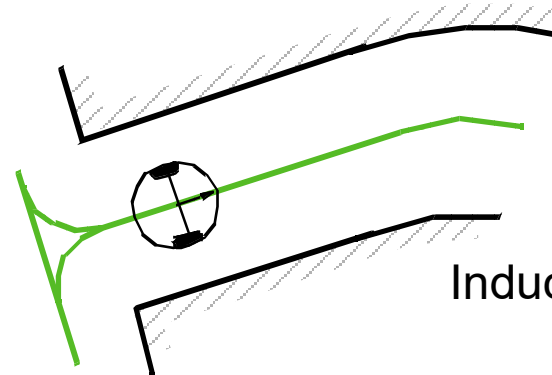
- Incrementally
(dead reckoning)



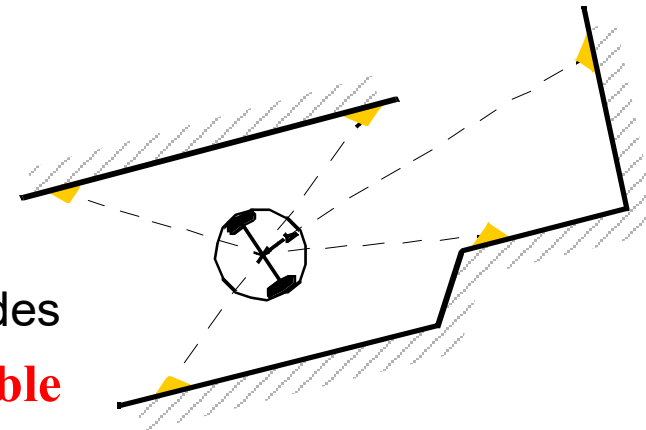
Odometric or inertial
sensors (gyro)

➤ *not applicable*

- Modifying the environments
(artificial landmarks / beacons)



Inductive or optical tracks (AGV)



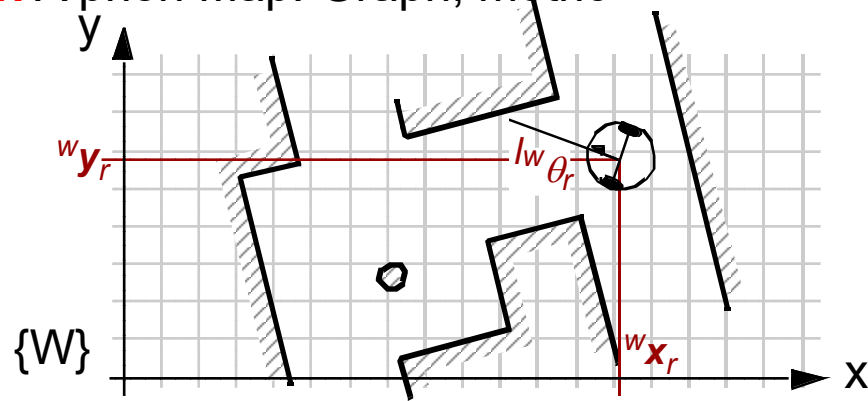
Reflectors or bar codes

- **expensive, inflexible**

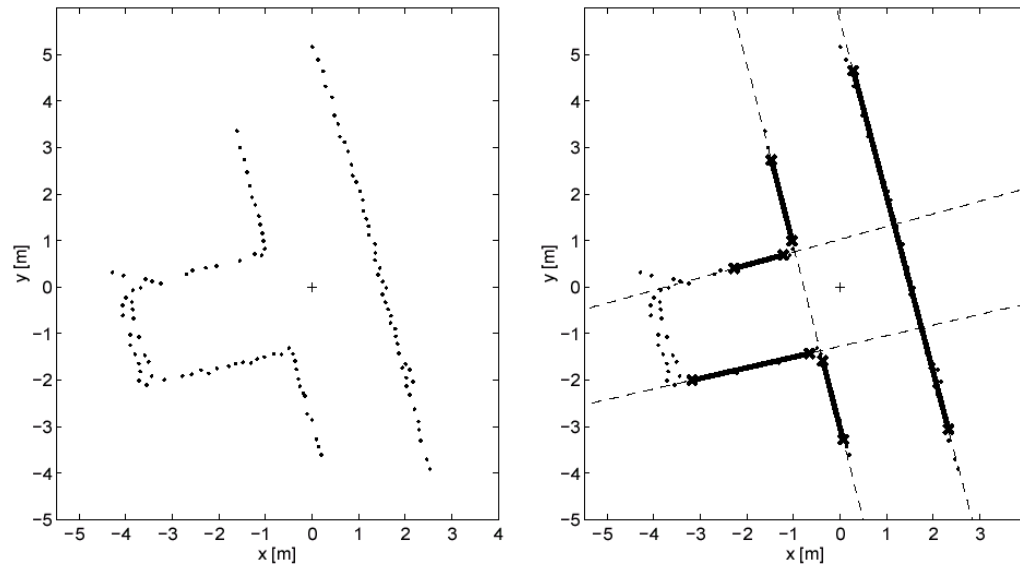


The Quantitative Metric Approach

1. A priori Map: Graph, metric

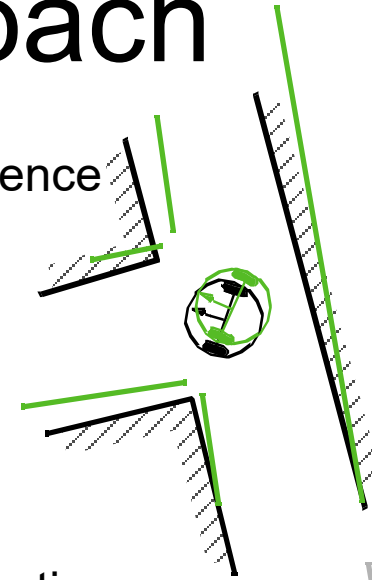


2. Feature Extraction (e.g. line segments)



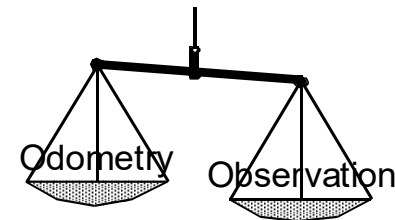
3. Matching:

Find correspondence of features



4. Position Estimation:

e.g. Kalman filter, Markov



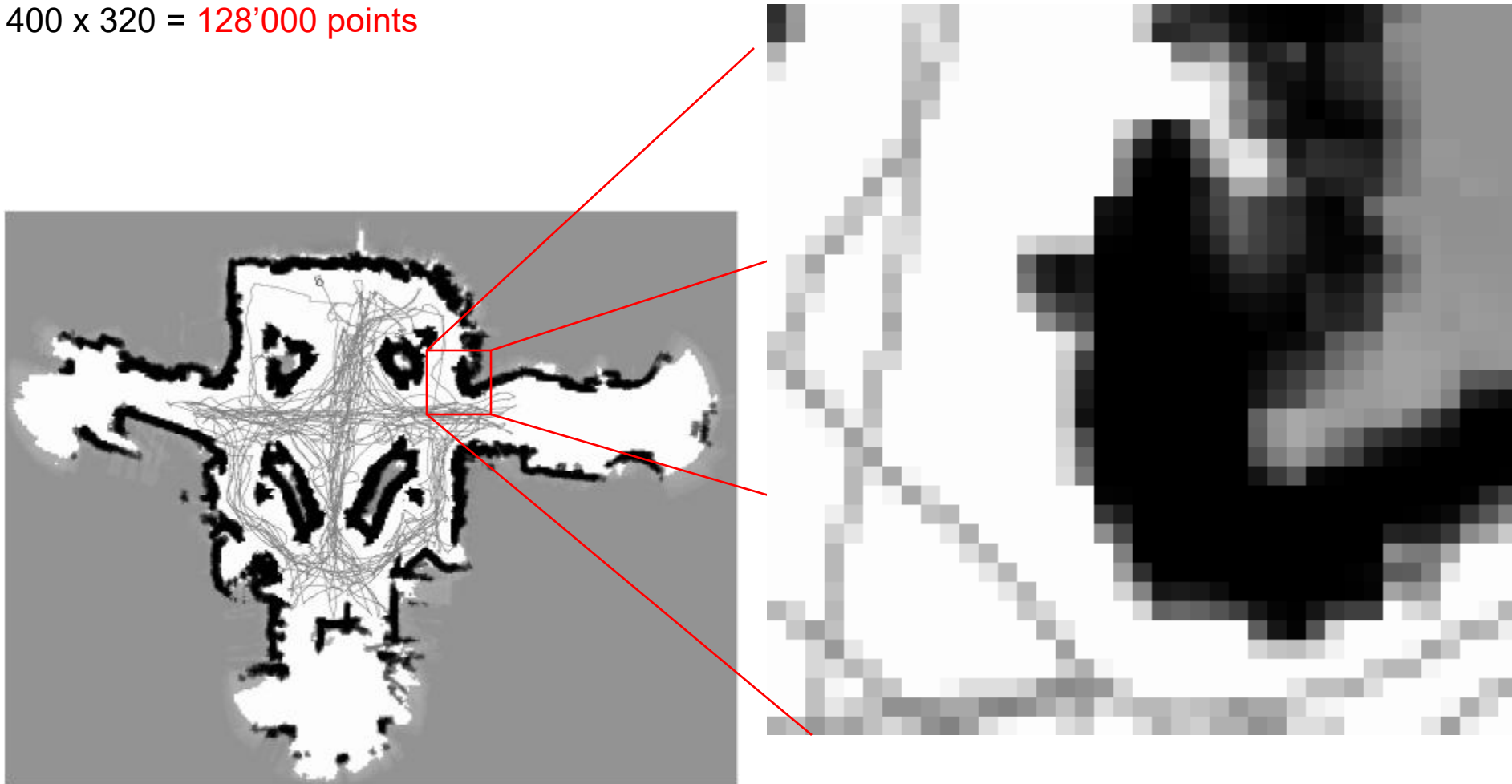
- representation of uncertainties
- optimal weighting acc. to a priori statistics



Grid-Based Metric Approach

- Grid Map of the Smithsonian's National Museum of American History in Washington DC

Grid: $\sim 400 \times 320 = 128'000$ points





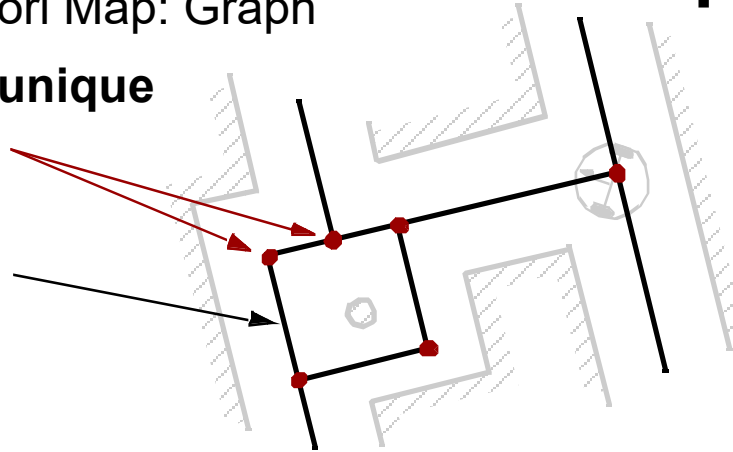
The Quantitative Topological Approach

1. A priori Map: Graph

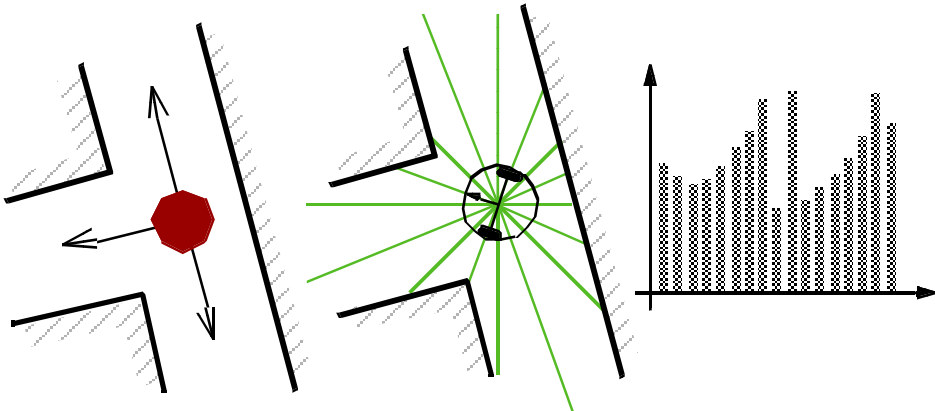
locally **unique**

points

edges



2. Method for determining the local uniqueness

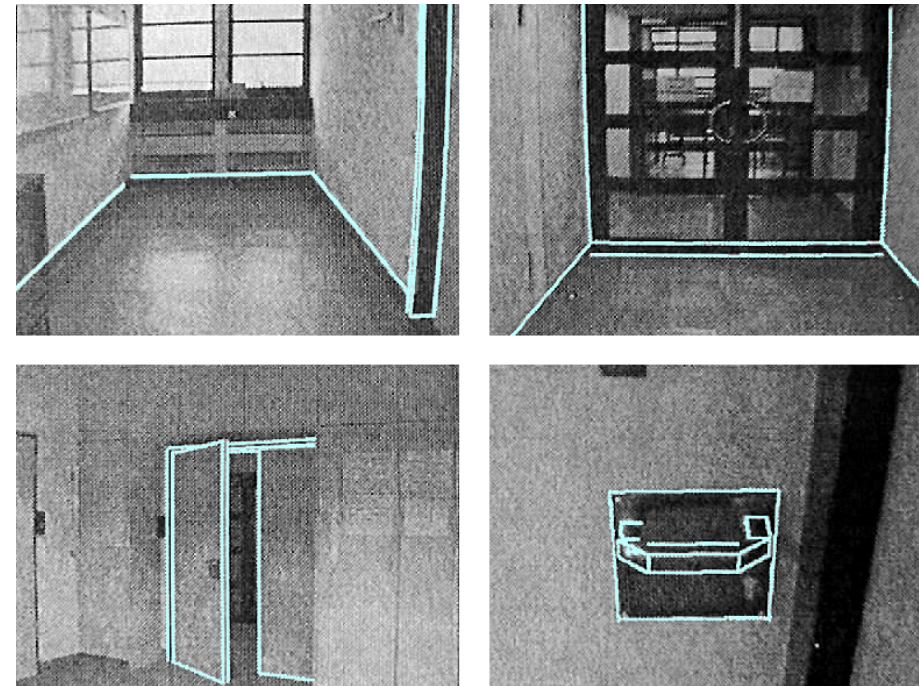


e.g. striking changes on raw data level
or highly distinctive features

3. Library of driving behaviors

e.g. wall or midline following, blind step,
enter door, application specific
behaviors

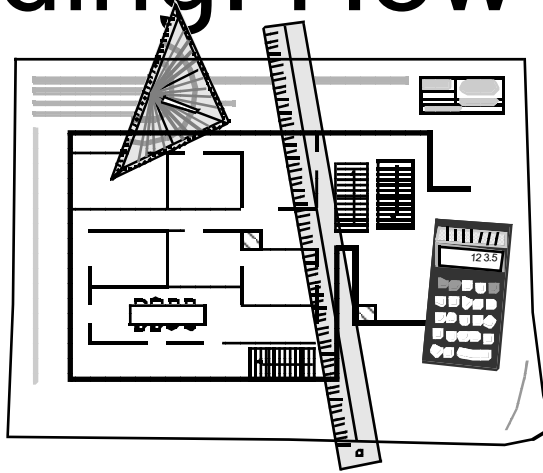
Example: Video-based navigation with
natural landmarks





Map Building: How to Establish a Map

1. By Hand



2. Automatically: Map Building

The robot **learns** its environment

Motivation:

- by hand: hard and costly
- dynamically changing environment
- different look due to different perception

3. Basic Requirements of a Map:

- a way to incorporate *newly sensed* information into the existing world model
- information and procedures for *estimating the robot's position*
- information to do *path planning* and other *navigation task* (e.g. obstacle avoidance)



predictability

▣ Measure of Quality of a map

- *topological correctness*
- *metrical correctness*

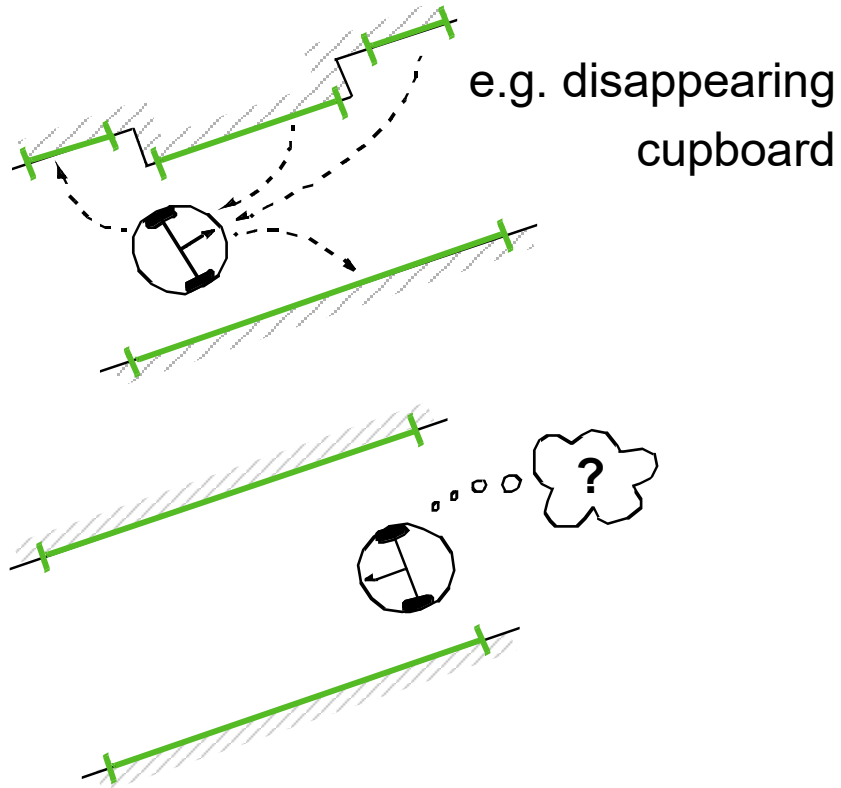
- ▣ But: Most environments are a mixture of *predictable* and *unpredictable* features
→ hybrid approach

model-based vs. behaviour-based



Map Building: The Problems

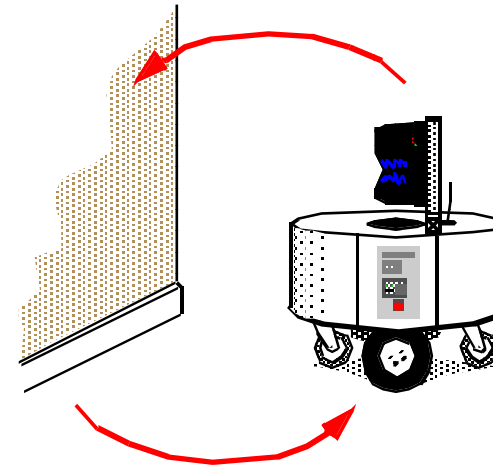
1. Map Maintaining: Keeping track of changes in the environment



- e.g. measure of **belief** of each environment feature

2. Representation and Reduction of Uncertainty

position of robot -> position of wall



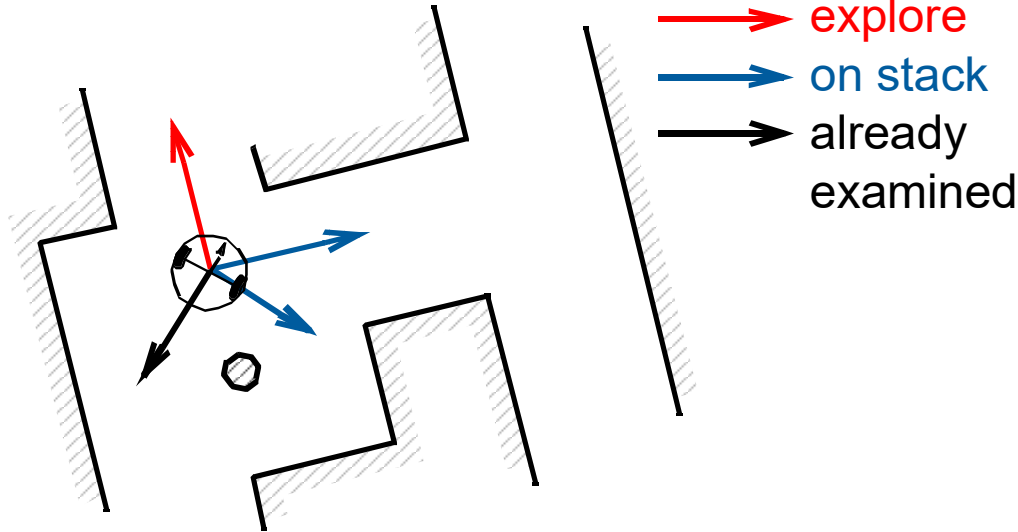
position of wall -> position of robot

- ▢ probability densities for feature positions
- ▢ additional exploration strategies



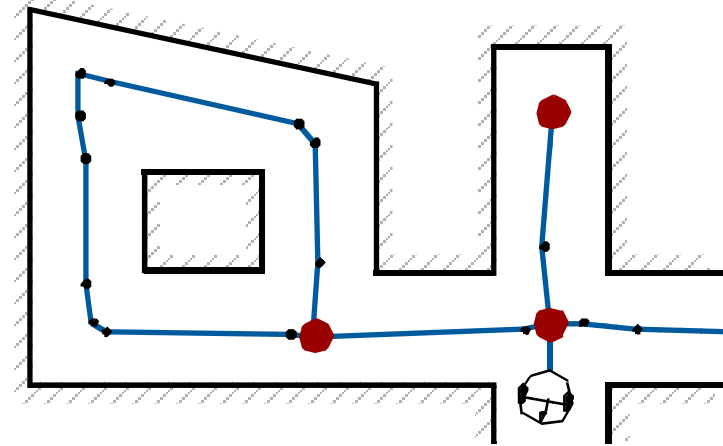
Exploration and Graph Construction

1. Exploration



- provides correct topology
- must recognize **already visited location**
- backtracking for unexplored openings

2. Graph Construction

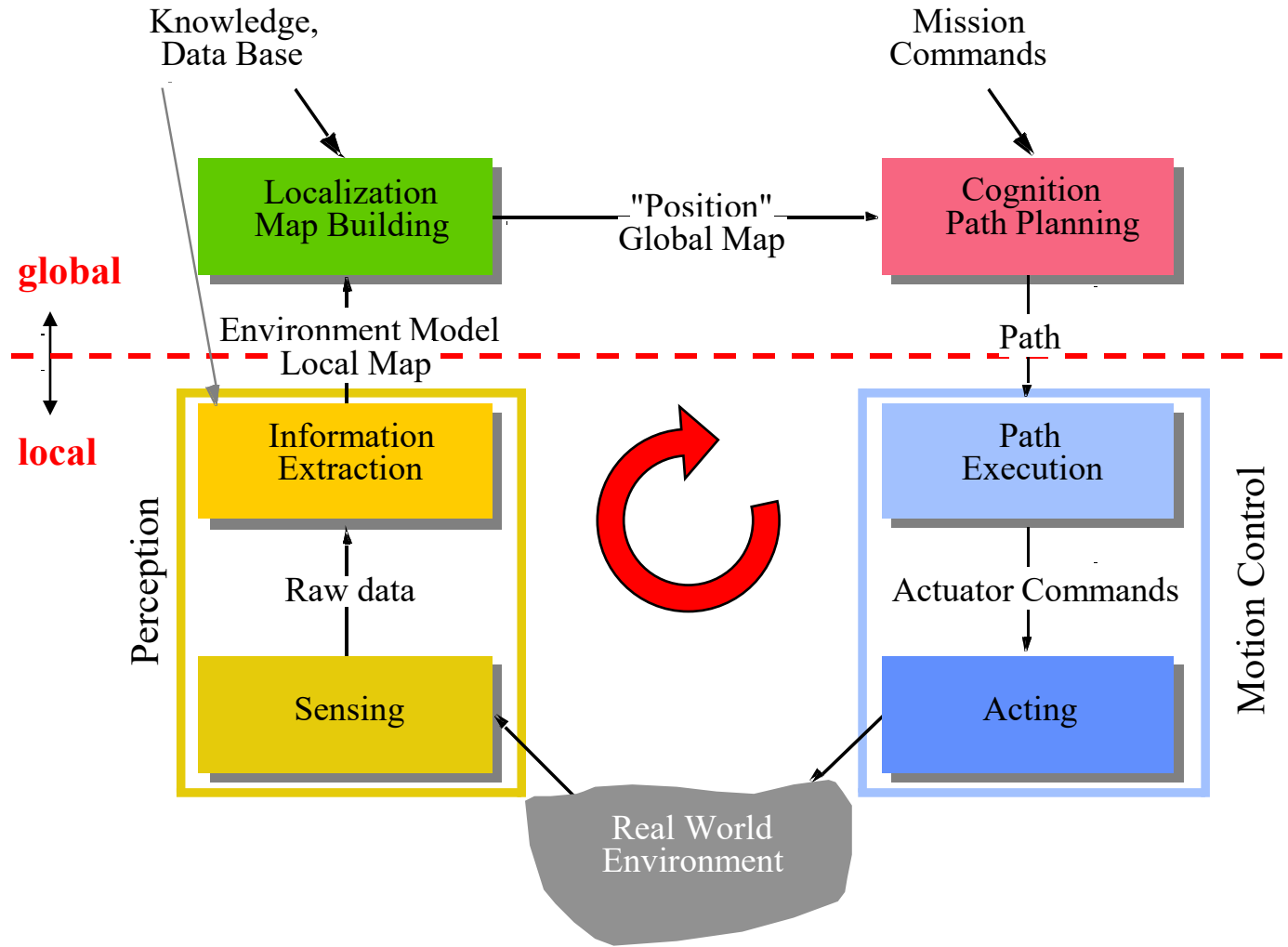


Where to put the **nodes**?

- Topology-based: at **distinctive locations**
- Metric-based: **where features disappear or get visible**



Control of Mobile Robots



- Most functions for *save navigation* are '*local*' not involving localization nor cognition
- Localization and global path planning
➔ slower update rate, only when needed
- This approach is pretty *similar* to what *human beings do*.



Locomotion

- A mobile robot requires locomotion mechanisms that allow it to move unbounded within its environment.
- A wide variety of locomotion methods exist, making locomotion selection an important design decision.
- Research robots have demonstrated diverse modes of movement, including walking, jumping, running, sliding, skating, swimming, flying, and rolling.
- Many robotic locomotion mechanisms are inspired by their biological counterparts



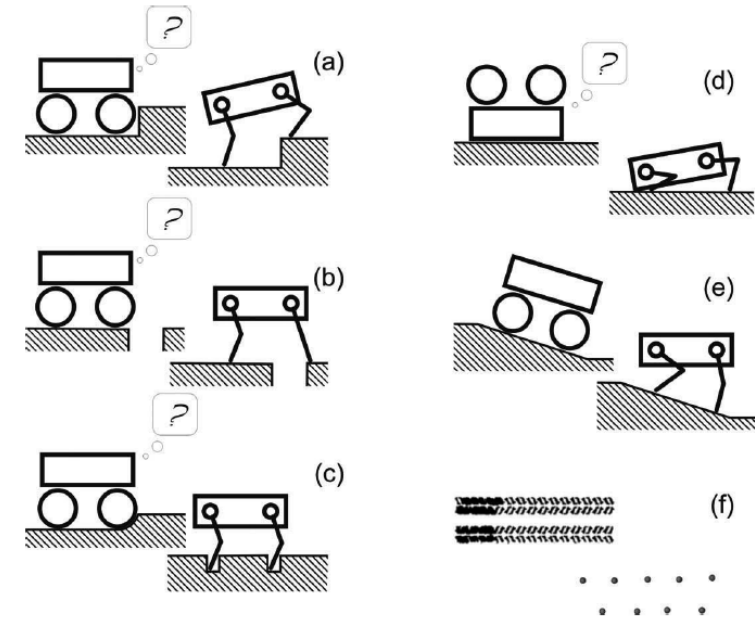
Locomotion

- Biological systems are highly effective at navigating harsh and unstructured environments.
- It is desirable to emulate biological locomotion mechanisms in mobile robot design.
- However, replicating natural locomotion is extremely difficult due to factors such as mechanical complexity.
- Biological muscles and hydraulic systems provide superior torque, response time, and efficiency compared to engineered systems of similar scale.
- As a result, mobile robots typically use:
 - **Wheeled mechanisms**, or
 - **A small number of articulated legs**, the simplest biological analogues.



Mobile Robots with legs

- Legged robots are particularly suited for rough terrain
 - (a) steps
 - (b) gaps
 - (c) sandy patches that are impassable for wheeled systems.
- The high number of degrees of freedom allows the robot to stand up when fallen (d) and keep its payload leveled (e).
- Because legged systems do not require a continuous path for support, they can rely on a few selected footholds, which also reduces the environmental impact (f).





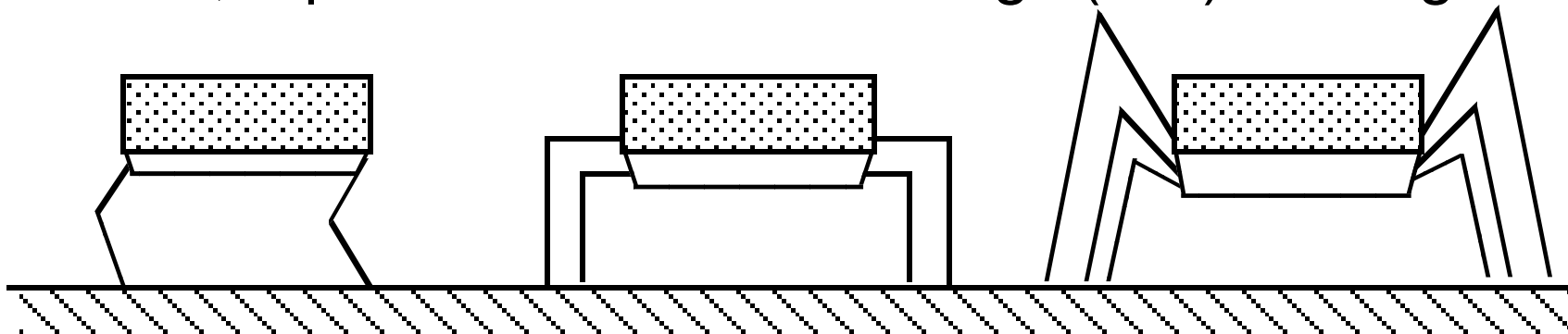
Key issues for locomotion

- Stability
 - Number and geometry of contact points
 - Center of gravity
 - Static/dynamic stability
 - Inclination of terrain
- Characteristics of contact
 - Contact point/path size and shape
 - Angle of contact
 - Friction
- Type of environment
 - Structure
 - Medium (e.g., water, air, soft or hard ground)



Mobile Robots with legs (walking machines)

- The fewer legs the more complicated becomes locomotion
 - stability, at least three legs are required for static stability
- During walking some legs are lifted
 - Thus, losing stability?
- For static walking at least 6 legs are required
 - babies have to learn for quite a while until they are able to stand or even walk on there two legs.
- Mammal, reptiles and insects four legs (two) four legs six legs





Walking Robots with Four Legs (Quadruped)

- Artificial Dog Aibo from Sony, Japan





Walking Robots with Four Legs (Quadruped)

- Spot Mini from Boston Dynamics





Walking Robots with Two Legs (Bipedal)

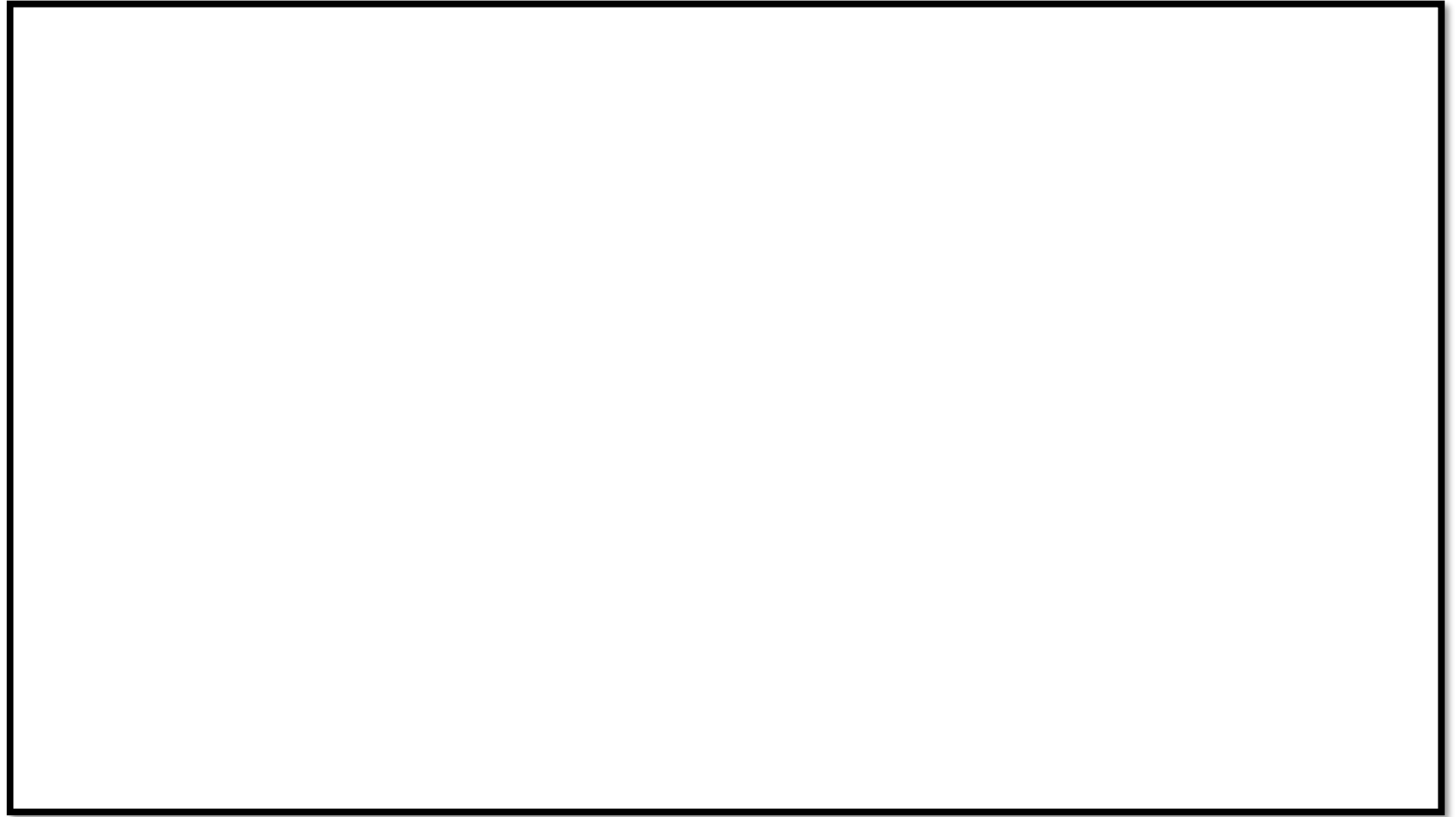
- H1 Humanoid Robot from Unitree





Walking Robots with Two Legs (Bipedal)

- Agility Robotics





Mobile Robots with Wheels

- Wheels are the most appropriate solution for most applications
- Three wheels are sufficient and to guarantee stability
- With more than three wheels a flexible suspension is required
- Selection of wheels depends on the application



Example of Wheeled Mobile Robots





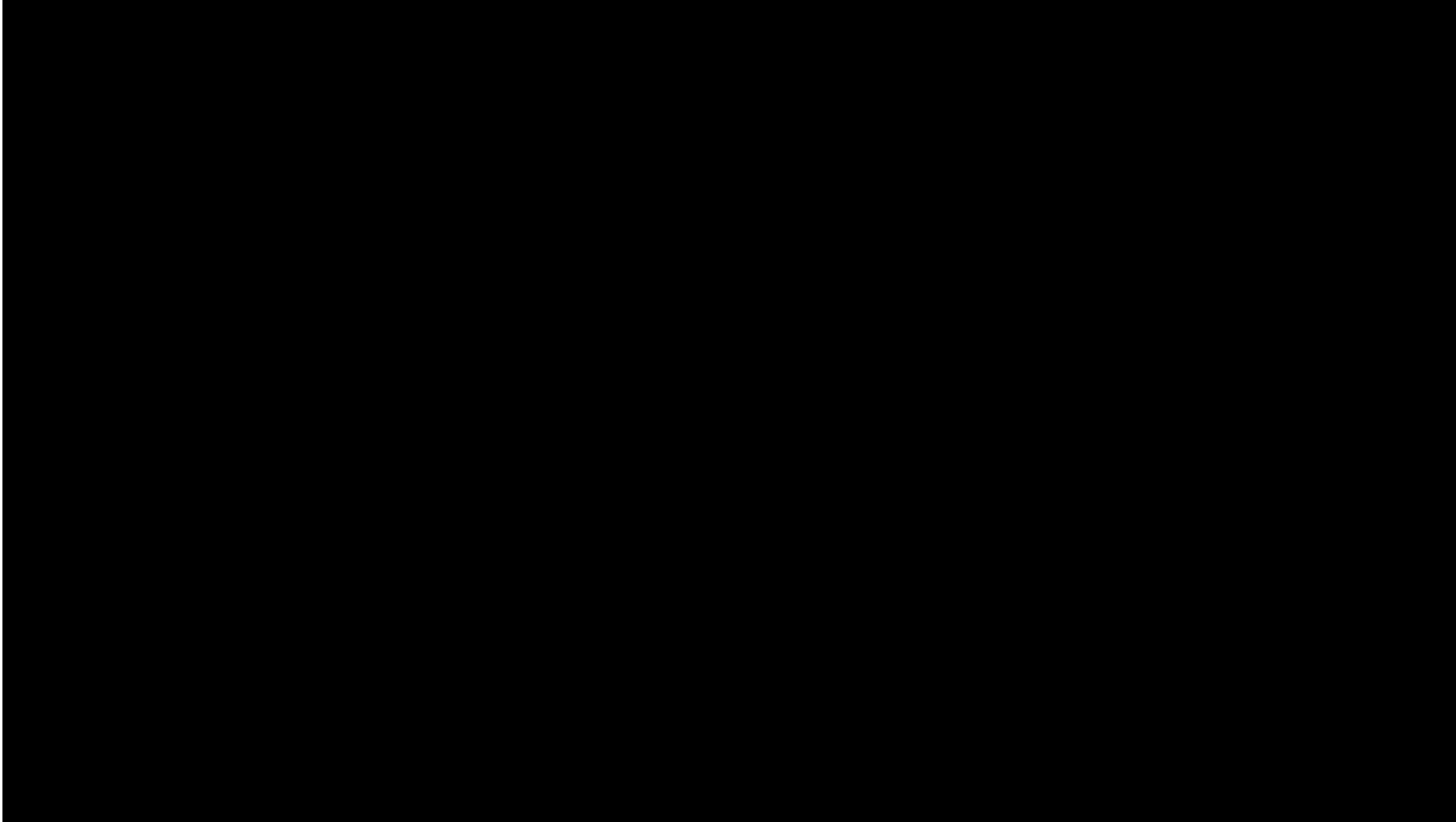
Example of Wheeled Mobile Robots

Time lapse
photograph of a
Roomba robot
cleaning a room





Hypered legged-Wheeled Robots



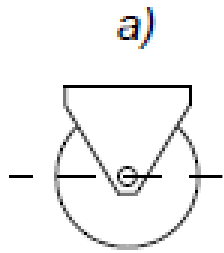


Wheeled Mobile Robots

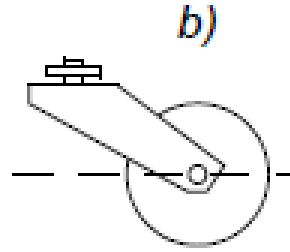
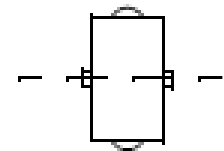
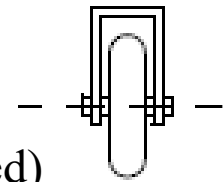
- The **wheel** has been by far the most widely used locomotion mechanism in both mobile robotics and man-made vehicles.
- **Balance is generally guaranteed** in wheeled robots, as they are typically designed so that all wheels remain in continuous contact with the ground.
- **Three wheels are sufficient** to ensure static stability; however, **two-wheeled robots** can also achieve stability through active control.
- When **more than three wheels** are used, a **suspension system** is required to maintain ground contact on uneven terrain.
- Research in wheeled mobile robotics primarily focuses on:
 - **Traction and stability**
 - **Maneuverability**
 - **Control**



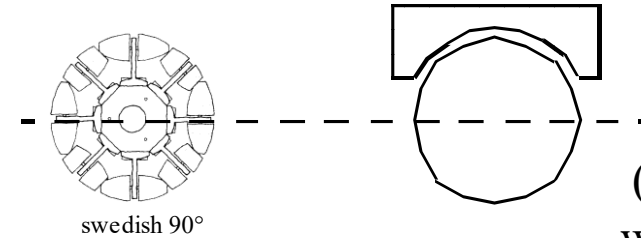
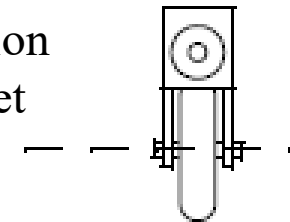
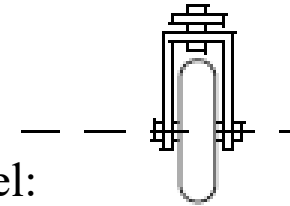
Wheel design: The Four Basic Wheel Types



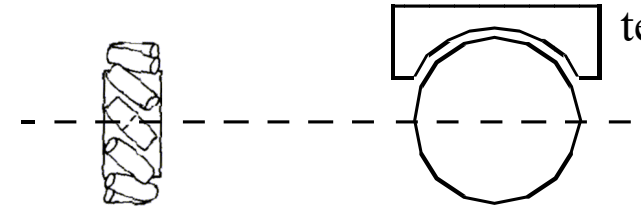
(a) Standard wheel:
two degrees of
freedom; rotation
around the (motorized)
wheel axle and the
contact point.



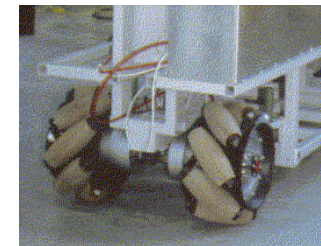
(b) castor wheel:
two degrees of
freedom; rotation
around an offset
steering joint.



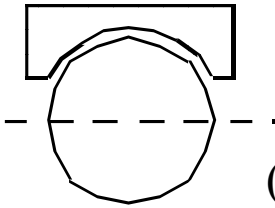
swedish 90°



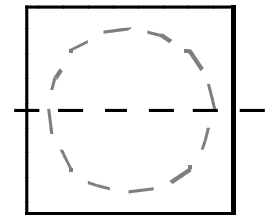
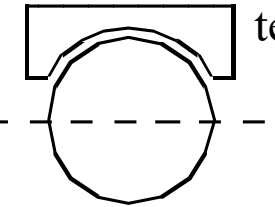
swedish 45°



(c) Swedish wheel: three degrees of freedom;
rotation around the (motorized) wheel axle, around
the rollers, and around the contact point.



(d) Ball or spherical
wheel: realization
technically difficult





Wheel Design

- A **standard wheel** allows steering without side effects because its center of rotation passes through the contact point with the ground.
- A **castor wheel** rotates about an offset axis, introducing forces on the robot chassis during steering.
- An **omni-directional (Swedish) 90° wheel** is similar to a standard wheel but includes passive rollers around its circumference.
- The rollers enable **low lateral friction**, allowing omni-directional motion while the wheel is driven normally.



Omni-directional
(or Swedish) 90°
wheel



Wheel Geometry (Wheel Arrangement)

- The choice of wheel types for a mobile robot is strongly linked to the choice of wheel arrangement, or wheel geometry.
- The designer must consider these two issues simultaneously when designing the locomotion mechanism of a wheeled robot.
- Unlike automobiles, mobile robots are designed for applications in a wide variety of situations.
- Automobiles all share similar wheel configurations because there is one design that maximizes maneuverability, controllability, and stability for their standard environment.



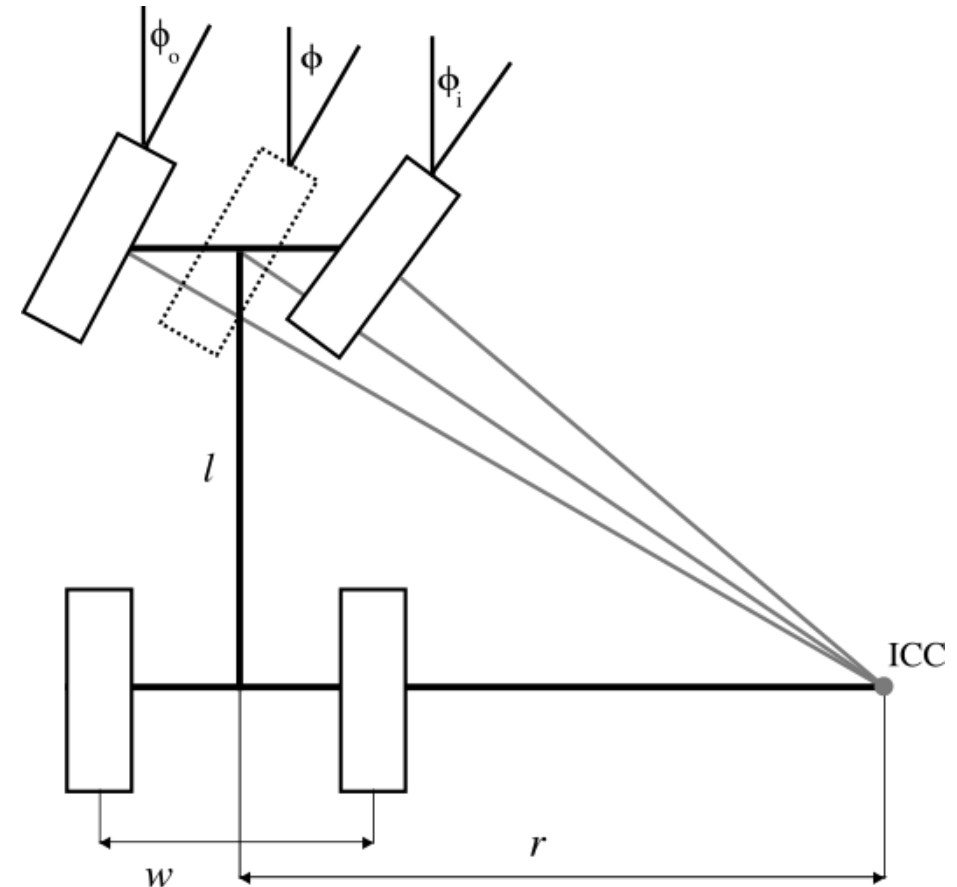
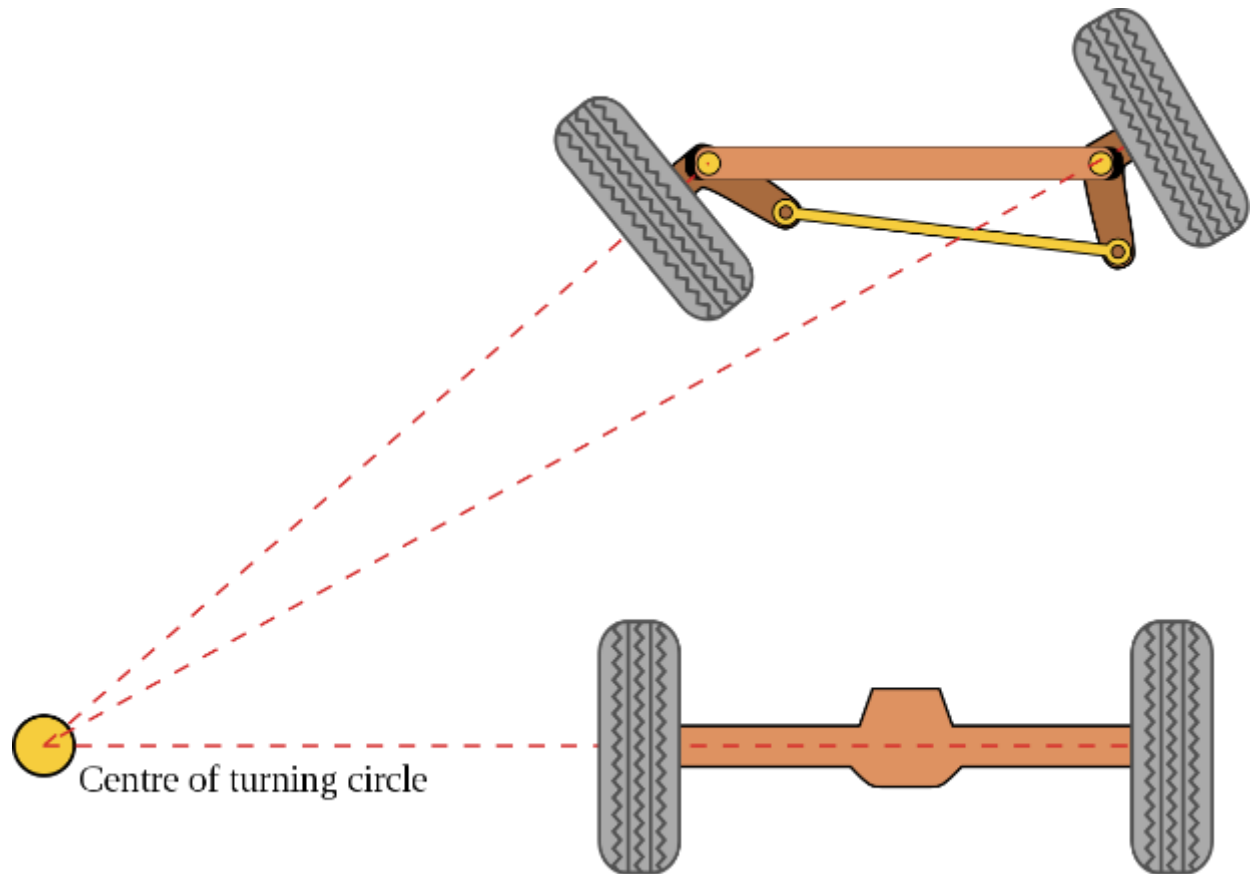
Wheel Geometry (Wheel Arrangement)

- There is no single wheel configuration that maximizes **Maneuverability**, **Controllability** and **Stability** for the variety of environments faced by different mobile robots.
- There is a great variety in the wheel configurations of mobile robots.
- In fact, few robots use the **Ackerman wheel configuration** for mobile robots designed for the road system



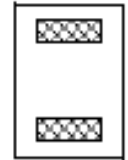
Wheel Geometry (Wheel Arrangement)

- Ackerman steering has poor maneuverability.





Wheel Geometry (Wheel Arrangement)

# of wheels	Arrangement	Description	Typical examples
2		One steering wheel in the front, one traction wheel in the rear	Bicycle, motorcycle
		Two-wheel differential drive with the center of mass (COM) below the axle	Cye personal robot

Icons for the each wheel type are as follows:	
	unpowered omnidirectional wheel (spherical, castor, Swedish)
	motorized Swedish wheel (Stanford wheel)
	unpowered standard wheel
	motorized standard wheel
	motorized and steered castor wheel
	steered standard wheel
	connected wheels

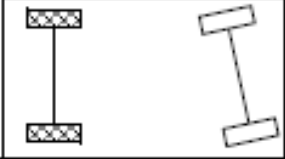
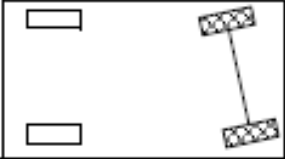
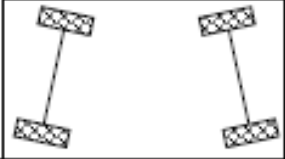


Wheel Geometry (Wheel Arrangement)

3		Two-wheel centered differential drive with a third point of contact	Nomad Scout, smartRob EPFL
		Two independently driven wheels in the rear/front, one unpowered omnidirectional wheel in the front/rear	Many indoor robots, including the EPFL robots Pygmalion and Alice
		Two connected traction wheels (differential) in rear, one steered free wheel in front	Piaggio minitrucks
		Two free wheels in rear, one steered traction wheel in front	Neptune (Carnegie Mellon University), Hero-1
		Three motorized Swedish or spherical wheels arranged in a triangle; omnidirectional movement is possible	Stanford wheel Tribolo EPFL, Palm Pilot Robot Kit (CMU)
		Three synchronously motorized and steered wheels; the orientation is not controllable	"Synchro drive" Denning MRV-2, Georgia Institute of Technology, I-Robot B24, Nomad 200

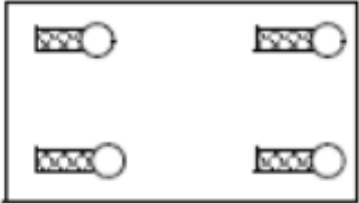


Wheel Geometry (Wheel Arrangement)

# of wheels	Arrangement	Description	Typical examples
4		Two motorized wheels in the rear, two steered wheels in the front; steering has to be different for the two wheels to avoid slipping/skidding.	Car with rear-wheel drive
		Two motorized and steered wheels in the front, two free wheels in the rear; steering has to be different for the two wheels to avoid slipping/skidding.	Car with front-wheel drive
		Four steered and motorized wheels	Four-wheel drive, four-wheel steering Hyperion (CMU)
		Two traction wheels (differential) in rear/front, two omnidirectional wheels in the front/rear	Charlie (DMT-EPFL)

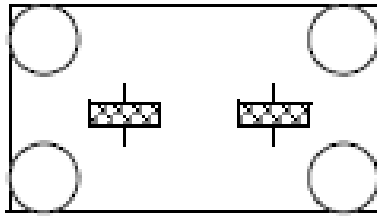
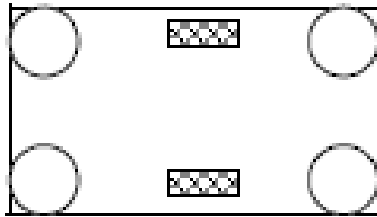


Wheel Geometry (Wheel Arrangement)

# of wheels	Arrangement	Description	Typical examples
4		Four omnidirectional wheels	Carnegie Mellon Uranus
		Two-wheel differential drive with two additional points of contact	EPFL Khepera, Hyperbot Chip
		Four motorized and steered castor wheels	Nomad XR4000



Wheel Geometry (Wheel Arrangement)

# of wheels	Arrangement	Description	Typical examples
6		Two motorized and steered wheels aligned in center, one omnidirectional wheel at each corner	First
		Two traction wheels (differential) in center, one omnidirectional wheel at each corner	Terregator (Carnegie Mellon University)



Stability

- Surprisingly, the minimum number of wheels required for static stability is two.
- A two-wheel differential-drive robot can achieve static stability if the center of mass is below the wheel axle.
- Usually, this requires wheel diameters that are impractically large.
- Also, dynamics can also cause a two-wheeled robot to strike the floor with a third point of contact, for instance, with sufficiently high motor torques from standstill.



Stability

- Conventionally, static stability requires a minimum of three wheels, **with the additional caveat that the center of gravity must be contained within the triangle formed by the ground contact points of the wheels.**
- Stability can be further improved by adding more wheels, although once the number of contact points exceeds three, the hyper-static nature of the geometry will require some form of **flexible suspension on uneven terrain.**



Stability

- The Segway PT is a two-wheeled, self balancing transportation (dynamic stability).
- Computers and motors in the base of the device keep the Segway PT upright.
- A user commands the Segway to go forward by shifting their weight forward on the platform, and backward by shifting their weight backward.
- The Segway notices, as it balances, the change in its center of mass, and first establishes and then maintains a corresponding speed, forward or backward.
- Gyroscopic sensors and fluid-based leveling sensors are used to detect the shift of weight. To turn, the user manipulates the handlebar left or right.



[The Reason Why Google Glass, Amazon Fire Phone and Segway All Failed \(forbes.com\)](http://forbes.com)



Maneuverability

- Some robots are omnidirectional, meaning that they can move at any time in any direction along the ground plane regardless of the orientation of the robot around its vertical axis.
- This level of maneuverability requires wheels that can move in more than just one direction, and so omnidirectional robots usually employ Swedish or spherical wheels that are powered.
- The four-caster wheel configuration in which each castor wheel is actively steered and actively translated is truly omnidirectional configuration.
- With a circular chassis and an axis of rotation at the center of the robot, such a robot can spin without changing its ground footprint.



Maneuverability

- The most popular such robot is the two-wheel differential-drive robot where the two wheels rotate around the center point of the robot.
- One or two additional ground contact points may be used for stability, based on the application specifics.
- The Ackerman steering configuration common in automobiles has a turning diameter that is larger than the car.
- Furthermore, for such a vehicle to move sideways requires a parking maneuver consisting of repeated changes in direction forward and backward.
- In addition, the limited maneuverability of Ackerman steering has an important advantage: its directionality and steering geometry provide it with **very good lateral stability in high-speed turns**.



Controllability

- There is generally an inverse correlation between controllability and maneuverability.
- For example, the omnidirectional designs such as the four-caster wheel configuration require significant processing to convert desired rotational and translational velocities to individual wheel commands.
- Furthermore, such omnidirectional designs often have greater degrees of freedom at the wheel.
- For instance, the Swedish wheel has a set of free rollers along the wheel perimeter.
- These degrees of freedom cause an accumulation of slippage, tend to reduce dead-reckoning accuracy and increase the design complexity.



Controllability

- Controlling an omnidirectional robot for a specific direction of travel is also more difficult and often less accurate when compared to less maneuverable designs.
- For example, an Ackerman steering vehicle can go straight simply by locking the steerable wheels and driving the drive wheels.
- In a differential-drive vehicle, the two motors attached to the two wheels must be driven along exactly the same velocity profile, which can be challenging considering variations between wheels, motors, and environmental differences.



Controllability

- With four-wheel omni-drive, the problem is even harder because all four wheels must be driven at exactly the same speed for the robot to travel in a perfectly straight line.
- The relationship between the individual wheel velocities (v_0, v_1, v_2, v_3) and the robot motion is given by:

$$\begin{aligned}v_x &= (v_0 + v_1 + v_2 + v_3)/4 \\v_y &= (-v_0 + v_1 + v_2 - v_3)/4 \\v_\theta &= (-v_0 + v_1 - v_2 + v_3)/4 \\v_{\text{error}} &= v_0 + v_1 - v_2 - v_3/4 \approx 0\end{aligned}$$





Controllability

- There is no “ideal” drive configuration that simultaneously maximizes stability, maneuverability, and controllability.
- Each mobile robot application places unique constraints on the robot design problem, and the designer’s task is to choose the most appropriate drive configuration .

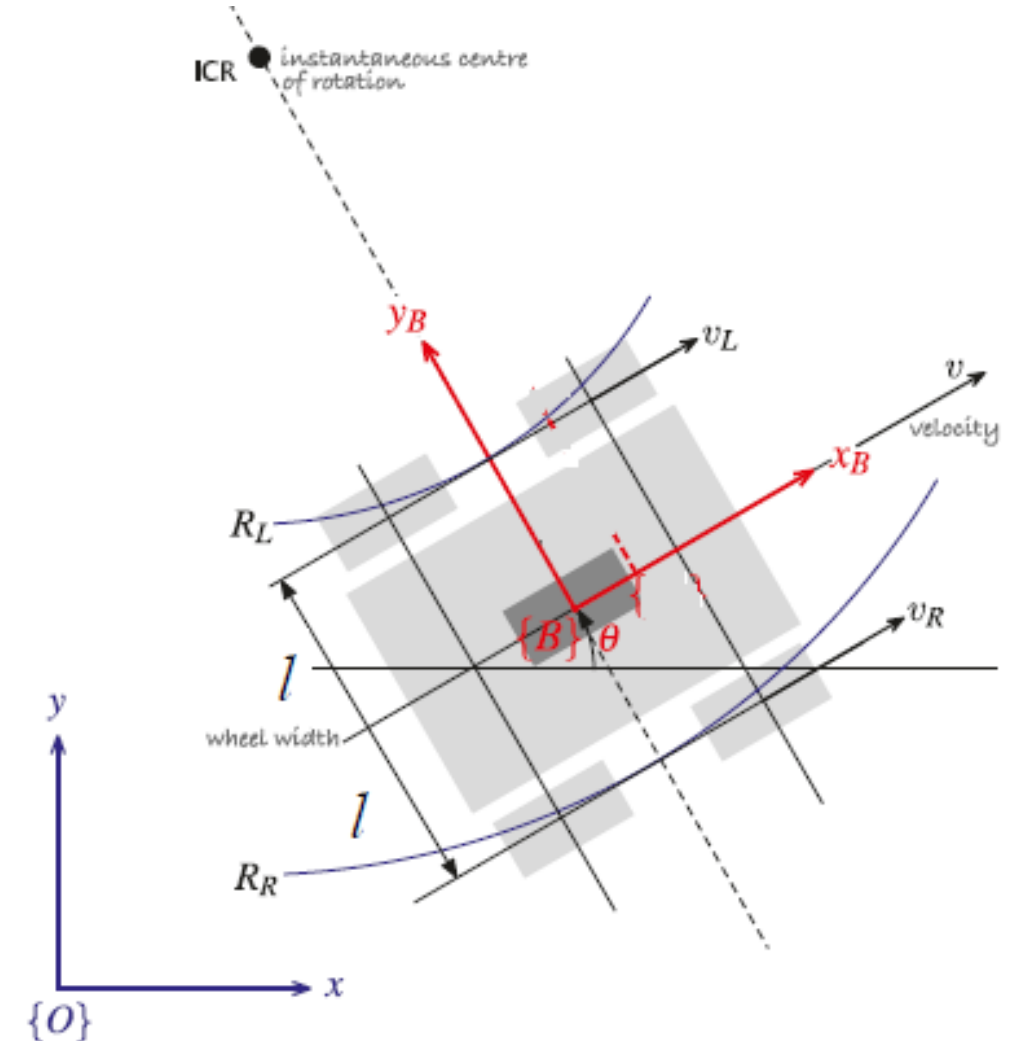


differential-drive robot

$$\begin{aligned}\dot{\theta} &= \frac{v_L}{R_L} = \frac{v_R}{R_R} \\ \dot{x} &= \frac{v_L + v_R}{2} \cos \theta \\ \dot{y} &= \frac{v_L + v_R}{2} \sin \theta \\ \dot{\theta} &= \frac{v_R - v_L}{2l}\end{aligned}$$

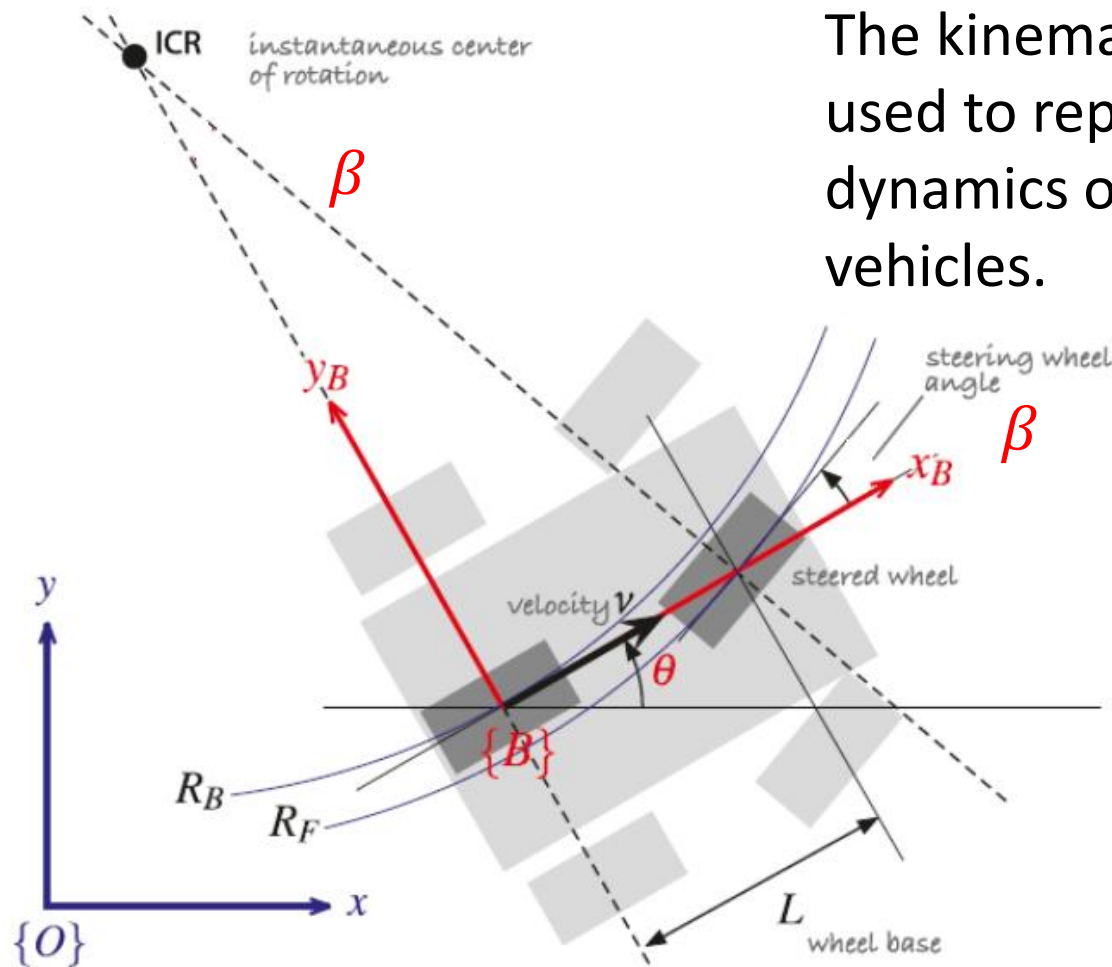
In terms of $\dot{\phi}_R$ and $\dot{\phi}_L$

$$\begin{aligned}\dot{x} &= \frac{r_l \dot{\phi}_L + r_r \dot{\phi}_R}{2} \cos \theta \\ \dot{y} &= \frac{r_l \dot{\phi}_L + r_r \dot{\phi}_R}{2} \sin \theta \\ \omega = \dot{\theta} &= \frac{r_r \dot{\phi}_R - r_l \dot{\phi}_L}{2l}\end{aligned}$$





Car-Like Mobile Robots



The kinematic bicycle model is widely used to represent the low-speed dynamics of four-wheeled car-like vehicles.



Car-Like Mobile Robots

- For Back-drive Car

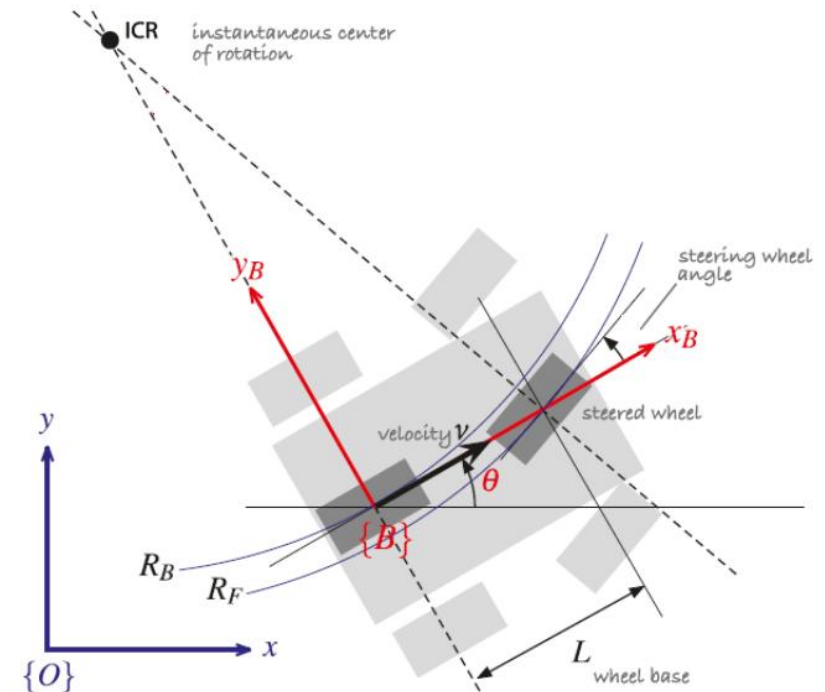
$$v = r\dot{\phi}$$

$$\dot{X}_R = v = r\dot{\phi}$$

$$\dot{Y}_R = 0$$

$$\omega = \frac{r\dot{\phi}}{L} \tan \beta$$

$$\dot{\xi}_I = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ \frac{\tan \beta}{L} \end{bmatrix} r\dot{\phi}$$





Car-Like Mobile Robots

- Kinematic model describes the velocities of the vehicle but not the forces or torques that cause the velocity.
- The rate of change of heading $\dot{\theta}$ is referred to as turn rate, heading rate or yaw rate.
- $\dot{\theta}$ can be measured by a gyroscope.
- When $v = 0$ then $\dot{\theta} = 0$; that is, it is not possible to change the vehicle's orientation when it is not moving.
- We must be moving to turn.
- When the steering angle $\beta = \frac{\pi}{2}$ the front wheel is orthogonal to the back wheel, the vehicle cannot move forward.



Any Questions